Container-based high-performance computing for modeling large-scale biological neural networks

Oliver James and **Sungho Hong**
Center for Cognition and Sociality, Institute for Basic Science

Introduction

1. Modeling large-scale, biophysically detailed neural networks increasingly relies on high-performance computing (HPC), but deploying such simulations across diverse supercomputing environments remains challenging due to software incompatibilities and complex dependencies

2. Singularity containers address these issues by encapsulating the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on shared HPC systems.

3. Despite these advantages, container adoption in computational neuroscience is still limited, and practical demonstrations—such as our cerebellar mossy fiber–granule cell network simulations—are needed to showcase their effectiveness for scalable, reproducible research.

Containerization for Scientific Computing

Our Computational Framework (N=27)

A slight change in mask contrast induces blindsight-like behavioral performance.

Case Study: Cerebellar Mossy Fiber–Granule Cell Network

There exists a *switch* in the spatio-temporal dynamics of stimulus representation, indicating the involvement of distinct neural mechanisms.

In visible conditions, robust dynamic neural representations in sensory and central areas suggest that this mechanisms may be sufficient to support initial stages of conscious awareness.
During invisible conditions, with weak dynamic coding, early stable neural representations of the stimulus emerge from the frontal cortex, implicating a potential role of frontal recurrent neural circuit for conscious awareness.

Performance Across HPC Environments

Source space analysis reveals the initiation of stable neural representations, progressing from frontal to visual cortices, during conditions of perceptual invisibility.

Reproducibility Portability

A five-cell mesoscopic canonical model is proposed for each region within the source space

/home/olive/Desktop/nature_neuroscience/Ent

/home/olive/Desktop/figs/home/olive/Desktop/figs/tex/Poster.pdf

(a) γ_{PYR}

(b) γ_{VIP+}

/home/olive/Desktop/figs/tex pdfip-eps-conver

(c) γ_{PV+}

Acknowledgment

This work was supported by IBS-R001-D2-2023-a00